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          ****************************************************
          *                                                  *
          *            STAAD.Pro                             *
          *            Version  2007    Build 04             *
          *            Proprietary Program of                *
          *            Research Engineers,  Intl.            *
          *            Date=    APR  1, 2013                 *
          *            Time=    19:53: 6                     *
          *                                                  *
          *       USER ID: Parsons Binckerhoff               *
          ****************************************************



     1. STAAD SPACE
INPUT FILE: 130222 VM1 STAIRCASE 4.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 26-NOV-12
     4. JOB NAME VM1 STAIRCASE
     5. END JOB INFORMATION
     6. INPUT WIDTH 79
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 1 -95.335 0 -20.85; 2 -95.335 0 -22.05; 3 -98.085 3 -20.85; 4 -98.085 3 -22.05
    10. 6 -99.585 0 -22.55; 8 -99.585 2 -22.55; 9 -99.585 3 -20.85
    11. 10 -99.585 3 -22.55; 11 -102.835 0 -20.85; 13 -102.835 3 -20.85
    12. 15 -102.835 4.6 -20.85; 17 -102.835 5.8 -20.85; 18 -102.835 5.8 -22.05
    13. 19 -99.585 3 -22.05; 20 -98.085 0 -22.55; 21 -98.085 2 -22.55
    14. 22 -98.085 3 -22.55
    15. MEMBER INCIDENCES
    16. 1 3 1; 2 4 2; 3 4 3; 6 9 3; 7 19 4; 9 8 6; 13 10 8; 14 10 19; 15 17 9
    17. 16 18 19; 17 13 11; 22 15 13; 26 17 15; 28 18 17; 30 9 8; 31 19 9; 32 21 20
    18. 33 22 21; 34 22 4; 35 3 21
    19. DEFINE MATERIAL START
    20. ISOTROPIC STEEL
    21. E 2.05E+008
    22. POISSON 0.3
    23. DENSITY 76.8195
    24. ALPHA 1.2E-005
    25. DAMP 0.03
    26. END DEFINE MATERIAL
    27. MEMBER PROPERTY JAPANESE
    28. 17 22 26 TABLE ST H250X250X9
    29. 9 13 32 33 TABLE ST H250X250X9
    30. 1 2 6 7 15 16 TABLE ST C250X90X9
    31. 3 14 28 30 31 34 35 TABLE ST H200X200X8
    32. CONSTANTS
    33. MATERIAL STEEL ALL
    34. BETA 180 MEMB 2 16
    35. BETA 90 MEMB 9 13 17 22 26 32 33
    36. SUPPORTS
    37. 1 2 PINNED
    38. 6 11 20 FIXED
    39. MEMBER RELEASE
    40. 15 16 END MY MZ
    41. 15 16 START MY MZ
    42. 3 START MY MZ
    43. 3 END MY MZ
    44. 6 START MY MZ
    45. 7 START MY MZ
    46. LOAD 1 LOADTYPE NONE  TITLE DL
    47. SELFWEIGHT Y -1 LIST 1 TO 3 6 7 9 13 TO 17 22 26 28 30 TO 35
    48. MEMBER LOAD
    49. *0.75KPA DEAD LOAD*
    50. 1 2 6 7 15 16 UNI GY -0.25
    51. LOAD 2 LOADTYPE NONE  TITLE LL
    52. *5KPA LIVE LOAD*
    53. MEMBER LOAD
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    54. 1 2 6 7 15 16 UNI GY -2.5
    55. LOAD 3 LOADTYPE NONE  TITLE WLX
    56. MEMBER LOAD
    57. 1 TO 3 9 13 TO 17 22 26 28 31 TO 34 UNI GX -0.24
    58. LOAD 4 LOADTYPE NONE  TITLE WLZ
    59. MEMBER LOAD
    60. 1 2 6 7 9 13 15 TO 17 22 26 32 33 UNI GZ -0.24
    61. LOAD 5 LOADTYPE NONE  TITLE NLX
    62. SELFWEIGHT X -0.015 LIST 1 TO 3 6 7 9 13 TO 17 22 26 28 30 TO 35
    63. MEMBER LOAD
    64. 1 2 6 7 15 16 UNI GX -0.004
    65. LOAD 6 LOADTYPE NONE  TITLE NLZ
    66. SELFWEIGHT Z -0.015 LIST 1 TO 3 6 7 9 13 TO 17 22 26 28 30 TO 35
    67. MEMBER LOAD
    68. 1 2 6 7 15 16 UNI GZ -0.004
    69. LOAD 7 LOADTYPE NONE  TITLE 1.0DL+1.0LL
    70. REPEAT LOAD
    71. 1 1.0 2 1.0
    72. LOAD 8 LOADTYPE NONE  TITLE 1.0DL+1.0LL+1.0WLX
    73. REPEAT LOAD
    74. 1 1.0 2 1.0 3 1.0
    75. LOAD 9 LOADTYPE NONE  TITLE 1.0DL+1.0LL+1.0WLZ
    76. REPEAT LOAD
    77. 1 1.0 2 1.0 4 1.0
    78. LOAD 10 LOADTYPE NONE  TITLE 1.0DL+1.0LL+1.0NLX
    79. REPEAT LOAD
    80. 1 1.0 2 1.0 5 1.0
    81. LOAD 11 LOADTYPE NONE  TITLE 1.0DL+1.0LL+1.0NLZ
    82. REPEAT LOAD
    83. 1 1.0 2 1.0 6 1.0
    84. LOAD 21 LOADTYPE NONE  TITLE 1.0DL+1.0LL-1.0 WLX
    85. REPEAT LOAD
    86. 1 1.0 2 1.0 3 -1.0
    87. LOAD 22 LOADTYPE NONE  TITLE 1.0DL+1.0LL-1.0 WLZ
    88. REPEAT LOAD
    89. 4 -1.0 1 1.0 2 1.0
    90. LOAD 23 LOADTYPE NONE  TITLE 1.0DL+1.0LL-1.0 NLX
    91. REPEAT LOAD
    92. 1 1.0 2 1.0 5 -1.0
    93. LOAD 24 LOADTYPE NONE  TITLE 1.0DL+1.0LL-1.0 NLZ
    94. REPEAT LOAD
    95. 6 -1.0 1 1.0 2 1.0
    96. *ULTIMATE LOAD COMBINATIONS
    97. LOAD 25 LOADTYPE NONE  TITLE 1.0DL+1.6LL
    98. REPEAT LOAD
    99. 1 1.0 2 1.6
   100. LOAD 26 LOADTYPE NONE  TITLE 1.4DL+1.6LL
   101. REPEAT LOAD
   102. 1 1.4 2 1.6
   103. LOAD 27 LOADTYPE NONE  TITLE 1.2DL+1.2LL+1.2WLX
   104. REPEAT LOAD
   105. 1 1.2 2 1.2 3 1.2
   106. LOAD 28 LOADTYPE NONE  TITLE 1.2DL+1.2LL+1.2WLZ
   107. REPEAT LOAD
   108. 1 1.2 2 1.2 4 1.2
   109. LOAD 29 LOADTYPE NONE  TITLE 1.2DL+1.2LL+1.2NLX
   110. REPEAT LOAD
   111. 1 1.2 2 1.2 5 1.2
   112. LOAD 30 LOADTYPE NONE  TITLE 1.2LL+1.2LL+1.2NLZ
   113. REPEAT LOAD
   114. 1 1.2 2 1.2 6 1.2
   115. LOAD 41 LOADTYPE NONE  TITLE 1.2DL+1.2LL-1.2 WLX
   116. REPEAT LOAD
   117. 1 1.2 2 1.2 3 -1.2
   118. LOAD 42 LOADTYPE NONE  TITLE 1.2DL+1.2LL-1.2WLZ
   119. REPEAT LOAD
   120. 4 -1.2 1 1.2 2 1.2
   121. LOAD 43 LOADTYPE NONE  TITLE 1.2DL+1.2LL-1.2 NLX
   122. REPEAT LOAD
   123. 1 1.2 2 1.2 5 -1.2
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 124. LOAD 44 LOADTYPE NONE  TITLE 1.2DL+1.2LL-1.2 NLZ
 125. REPEAT LOAD
 126. 6 -1.2 1 1.2 2 1.2
 127. PERFORM ANALYSIS


          P R O B L E M   S T A T I S T I C S
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   NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =    17/    20/     5

         SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

   ORIGINAL/FINAL BAND-WIDTH=    13/     4/     30 DOF
   TOTAL PRIMARY LOAD CASES =   25, TOTAL DEGREES OF FREEDOM =     78
   SIZE OF STIFFNESS MATRIX =        3 DOUBLE  KILO-WORDS
   REQRD/AVAIL. DISK SPACE  =     12.1/  20378.6 MB



 128. DEFINE ENVELOP
 129. 25 TO 30 41 TO 44 ENVELOP 1 TYPE STRESS
 130. 7 TO 11 21 TO 24 ENVELOP 2 TYPE SERVICEABILITY
 131. END DEFINE ENVELOP
 132. LOAD LIST 1 TO 11 21 TO 25
 133. *PRINT JOINT DISPLACEMENTS ALL
 134. LOAD LIST 26 TO 30 41 TO 44
 135. *PRINT MEMBER FORCES ALL
 136. LOAD LIST ALL
 137. PARAMETER 1
 138. CODE BS5950
 139. SGR 0 MEMB 1 TO 3 6 7 9 13 TO 17 22 26 28 31 TO 34
 140. *TRACK 2
 141. CHECK CODE ALL


                        STAAD.Pro CODE CHECKING - (BSI )
                        ***********************

 PROGRAM CODE REVISION V2.12_5950-1_2000


ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

MEMBER     TABLE       RESULT/   CRITICAL COND/     RATIO/     LOADING/
                         FX            MY             MZ       LOCATION
=======================================================================


      1 ST  C250X90X9    PASS     BS-4.8.3.3.1      0.194        26
                        0.32 C         0.10           7.52        -
      2 ST  C250X90X9    PASS     BS-4.8.3.3.1      0.178        42
                        0.68 C         0.84           4.37        -
      3 ST  H200X200X8   PASS     BS-4.8.2.2        0.008        26
                       12.32 T         0.00          -0.12        0.60
      6 ST  C250X90X9    PASS     BS-4.8.3.3.1      0.234        26
                        0.01 C         2.44           3.26        -
      7 ST  C250X90X9    PASS     BS-4.8.3.3.1      0.091        42
                        0.25 C         0.90           1.59        -
      9 ST  H250X250X9   PASS     ANNEX I.1         0.135        26
                       33.97 C         0.85          33.46        -
     13 ST  H250X250X9   PASS     ANNEX I.1         0.094        26
                       10.19 C         0.24          24.04        -
     14 ST  H200X200X8   PASS     BS-4.2.3-(Y)      0.035        26
                       21.66 T         0.00           2.38        0.00
     15 ST  C250X90X9    PASS     BS-4.8.3.3.1      0.209        42
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     16 ST  C250X90X9    PASS     BS-4.8.3.3.1      0.209        28
                        0.07 C         0.66           6.46        -
     17 ST  H250X250X9   PASS     ANNEX I.1         0.087        28
                       21.43 C         0.39          21.87        -
     22 ST  H250X250X9   PASS     ANNEX I.1         0.057        28
                       18.90 C         0.19          14.45        -
     26 ST  H250X250X9   PASS     ANNEX I.1         0.050        26
                       22.61 C         0.11          12.80        -
     28 ST  H200X200X8   PASS     BS-4.8.3.3.2      0.089        26
                        0.00           0.04          12.83        -
     30 ST  H200X200X8   PASS     ANNEX I.1         0.067        26
                       31.52 C         0.07           9.42        -
     31 ST  H200X200X8   PASS     BS-4.8.2.2        0.032        26
                       22.11 T         0.02          -2.76        1.20
     32 ST  H250X250X9   PASS     ANNEX I.1         0.162        26
                       29.30 C         5.06          31.27        -
     33 ST  H250X250X9   PASS     ANNEX I.1         0.082        26
                       16.41 C         0.60          20.20        -
     34 ST  H200X200X8   PASS     BS-4.8.2.2        0.060        26
                       12.86 T        -0.09           7.34        0.00
     35 ST  H200X200X8   PASS     ANNEX I.1         0.091        26
                       14.78 C         2.90           6.78        -


   ************** END OF TABULATED RESULT OF DESIGN **************


 142. FINISH


          *********** END OF THE STAAD.Pro RUN ***********

            **** DATE= APR  1,2013   TIME= 19:53: 7 ****

      ************************************************************
      *         For questions on STAAD.Pro, please contact       *
      *   Research Engineers Offices at the following locations  *
      *                                                          *
      *              Telephone                Email              *
      *  USA:      +1 (714)974-2500       support@bentley.com    *
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      *  UK        +44(1454)207-000       support@bentley.com    *
      *  NORWAY    +47 67 57 21 30        staad@edr.no           *
      *  SINGAPORE +65 6225-6158          support@bentley.com    *
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